

INTRODUCTION TO DOCKER VOLUMES

1. Introduction to Docker Volumes

▪ What is a Docker Volume?

- A Docker Volume is a persistent storage mechanism used by containers to store and share data.
- Volumes are managed by Docker and stored outside the container filesystem, making them ideal for storing:
 - Application data
 - Configuration files
 - Database files
 - Logs

▪ Why Use Volumes?

- **Data persistence:** Data remains even after the container is deleted.
- **Container independence:** Volumes can be shared between containers.
- **Better performance:** Especially on Linux, volumes perform better than bind mounts.
- **Safe upgrades:** Preserve important files across container upgrades or rebuilds.

2. Docker Volume Commands

Command	Description
<code>docker volume ls</code>	List all Docker volumes
<code>docker volume create <name></code>	Create a new volume
<code>docker volume inspect <name></code>	Show detailed info about a volume
<code>docker volume rm <name></code>	Remove a specific volume
<code>docker volume prune</code>	Remove all unused (dangling) volumes

3. Using Volumes with docker run

Attach a volume to a container using the `-v` or `--mount` flag

▪ Using `-v` (short syntax)

`docker run -d -v my_volume:/data nginx`

▪ Using `--mount` (recommended for clarity)

`docker run -d --mount source=my_volume,target=/data nginx`

4. Sharing Volumes Between Containers

Two or more containers can share a volume:

○ `docker run -d -v shared_volume:/app/data container1`

○ `docker run -d -v shared_volume:/app/data container2`

Changes made by one container will be visible to others.